

1 Basic Knowledge & SVCDE Methods for ODE

1.1 Linear Algebra

Matrix Calculate: $\det(kA) = k^n \det(A)$ || $\det(A^{-1}) = \det(A)^{-1}$ || $(AB)^* = B^* A^*$ || $\det(A) = \det(A^T)$.

Eigenvalue λ & Eigenvector $\mathbf{x}$: Def: $A\mathbf{x} = \lambda\mathbf{x}$. Method: ${}^1 \det(A - \lambda I) = 0 \rightarrow \lambda$. ${}^2 (A - \lambda I)\mathbf{x} = 0 \rightarrow \mathbf{x}$.

· **Algebraic Multiplicity (AM) of λ :** λ 重复出现的次数. $\geq$ **Geometric Multiplicity (GM) of λ :** # $\mathbf{x}$ under λ .

· **Generalised Eigenvectors:** If $AM > GM$: Using $\xi_\lambda = (A - \lambda I)\eta_1, \eta_{n-1} = (A - \lambda I)\eta_n$ for each ξ_λ . η_j are *generalised eigenvectors*.

1.2 Calculus & SVCDE Method for ODE

Some Defs: $\nabla F = (\partial_x F, \partial_y F, \partial_z F, \dots)$ || **Simply Connected:** 没有洞, 相互连接的区域. || ∂D : 区域 D 的边界.

Method of Integrating Factors: $y' + p(x)y = q(x)$. $\Rightarrow$ Integrating Factor: $\mu = e^{\int p(x)dx}$, $\mu y = \int q(x) \cdot \mu dx$

Homogeneous 2nd ODE form: $ay'' + by' + c = 0 \Rightarrow$ Calculate $ar^2 + br + c = 0$, ps: $\Delta = b^2 - 4ac$, roots: r_1, r_2 ps: $y'' \pm \lambda y = 0$ 见附录

· ${}^1 \Delta > 0, y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$; ${}^2 \Delta = 0, y = (C_1 + C_2 x)e^{rx}$; ${}^3 \Delta < 0, y = e^{\alpha x}[C_1 \cos(\beta x) + C_2 \sin(\beta x)]$ ps: $r_1, r_2 = \alpha \pm \beta i$

non-Homogeneous 2nd ODE form: $ay'' + by' + c = g(x) \Rightarrow y = y_h + y_p$ ps: y_h is sol for $g(x) = 0$.

· Method of Undetermined Coefficients: Let $A_n(x) = \sum_{i=0}^n a_i x^i$, $B_n(x) = \sum_{i=0}^n b_i x^i$, $P_n(x) = \sum_{i=0}^n p_i x^i$.

1. If $g(x) = P_n(x) \Rightarrow y_p = x^s A_n(x)$, $s := 0, (0 \neq r_1, r_2); 1, (0 = r_1 \text{ or } r_2); 2, (0 = r_1 = r_2)$.

2. If $g(x) = M e^{kx} \Rightarrow y_p = C x^s e^{kx}$, $s := 0, (k \neq r_1, r_2); 1, (k = r_1 \text{ or } r_2); 2, (k = r_1 = r_2)$.

3. If $g(x) = M \cos(kx) + N \sin(kx) \Rightarrow y_p = x^s (C_1 \cos(kx) + C_2 \sin(kx))$, $s := 0, (r_1, r_2 \neq ik); 1, (r_1, r_2 = ik)$.

4. If $g(x) =$ product from 1,2,3. $\Rightarrow y_p =$ corresponding product of y_p (x^s 只乘一次), $s :=$ 令 y_p 与 y_c 没有相同项. (≤ 2)

Euler Equation: General: $a_n x^n y^{(n)} + a_{n-1} x^{n-1} y^{(n-1)} + \dots + a_0 y = 0 \Rightarrow$ Seek Solution as: $y = x^\alpha \Rightarrow$ solution: $y = \sum_i c_i x^{\alpha_i}$

1.3 ODE / PDE Definitions

Def of DEs / ODEs / PDEs: Given independent variables $(x_1, \dots, x_n)$, functions $(y_1(x_{j_1}), \dots, y_p(x_{j_p}))$ and some derivatives (e.g. $\partial_s y_t$).

· **DEs:** $\mathcal{F}_i[y_r, \dots, \partial_{x_t} y_b, x_1, \dots, x_n] = 0$, $i = s, s-1, \dots, 1$. || **ODEs:** $n = 1$ (无偏导) **PDEs:** $n > 1$ (有偏导)

Def of Linear: Let L 去掉 $y_i^{(n)}$ 的函数 (e.g. $L = \frac{d^2}{dx^2} - x$ for $y'' - xy = x^2$) $\Rightarrow$ Linear if: $L[a_1 y_1 + a_2 y_2] = a_1 L[y_1] + a_2 L[y_2]$

1.4 (ODE) Uniqueness Theorem & Autonomous

Existence & Uniqueness Theorem (Real func): Given $\mathbf{y}'(t) = \mathbf{F}(t, \mathbf{y}^T)$, with $\mathbf{y}(t_0) = \mathbf{y}_0$. Let $R_i = [t_0 - \delta, t_0 + \delta] \times \prod_j [y_{0j} - \delta_j, y_{0j} + \delta_j]$.

· If $F_j, \partial_{y_i} F_j : R_i \rightarrow \mathbb{R}$ are continuous. $\Rightarrow \exists$ unique solution $\mathbf{y}$ in R'_i Let $R'_i = [t_0 - \varepsilon, t_0 + \varepsilon] \times \prod_j [y_{0j} - \varepsilon_j, y_{0j} + \varepsilon_j] \subseteq R_i$.

Autonomous: For $\mathbf{y}'(t) = \mathbf{f}(t, \mathbf{y}^T)$. If $\partial_x \mathbf{f} = \mathbf{0} \Rightarrow$ autonomous. or: $\mathbf{f}(t, \mathbf{y}^T) = \mathbf{f}(\mathbf{y}^T)$ (不含 t)

2 Linear ODEs

n-th ODE $\rightarrow$ 1st ODEs: For $y^{(n)} = F(t, y, y', \dots, y^{(n-1)})$. $\Rightarrow$ Let $x_i = y^{(i-1)}$. Then $x'_n = F(t, x_1, x_2, \dots, x_{n-1})$ which is 1st ODE.

Def of Linear 1st ODEs: $\mathbf{y}'(t) = P(t)\mathbf{y}(t) + \mathbf{g}(t)$ (*), P is matrix-valued function.

Homogeneous: $\mathbf{g}(t) = \mathbf{0}$.

Existence & Uniqueness Theorem: If $P_{ij}, \mathbf{g}$ of (*) continuous in $R, \mathbf{y}(t_0) = \mathbf{y}_0 \Rightarrow \exists$ unique solution $\mathbf{y}$ in R' $R' = [t_0 - \varepsilon, t_0 + \varepsilon] \subseteq R = [t_0 - \delta, t_0 + \delta]$

2.1 Homogeneous

Solve: $\mathbf{y}' = P\mathbf{y}$ || If P constant, $P = A$ || If possible: λ, ξ, η be eigenvalues, eigenvectors, generalised eigenvectors.

Principle of Superposition: If $\mathbf{y}_i$ are solutions $\Rightarrow \sum_i c_i \mathbf{y}_i$ also.

$S := \{\mathbf{y} | \mathbf{y}' = P\mathbf{y}\}$ is a linear space, $\dim(S) = n$ if P_{ij} continuous.

Wronskian at x : $W[\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(n)}](t) = \begin{vmatrix} y_1^{(1)}(t) & \dots & y_1^{(n)}(t) \\ \vdots & \ddots & \vdots \\ y_n^{(1)}(t) & \dots & y_n^{(n)}(t) \end{vmatrix}$.

$W(t) \neq 0, \forall t \Rightarrow \{\mathbf{y}^{(j)}\}$ **linearly independent**. ps: 经常用 Abel Theorem

· Geometric Meaning: Geometrically, W is the volume spanned by the three vectors $\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \mathbf{y}^{(3)}$; t 增加或减小代表在 $\mathbf{y}^{(i)}$ 方向上 compress/stretch.

Abel's Theorem: If $W(t_0) \neq 0, t_0 \in I \Rightarrow \forall t \in I, W(t) \neq 0$ || **Abel's Formula:** $W(t) = W(t_0) e^{\int_{t_0}^t \text{tr}[P(s)]ds}$; $t_0, t \in I$ $P(t)$ is continuous in I .

Fundamental Set of Solutions (FSoS): If $\{\mathbf{y}^{(j)}(t)\}$ linearly independent at $t \in I$, it is FSoS in I .

General Solution: If $\{\mathbf{y}^{(j)}(t)\}_{j=1}^n$ is FSoS (Linearly Independent) $\Rightarrow$ General solution: $\mathbf{y} = \sum_{j=1}^n c_j \mathbf{y}_j$

2.1.1 Eigenvalue/vector Method (A constant)

AM=GM|Real λ : general solution: $\mathbf{y} = \sum_j c_j e^{\lambda_j t} \xi_j$.

AM=GM|Complex λ : (ctd.) For $\lambda, \bar{\lambda} \in \mathbb{C}$. $\Rightarrow e^{\lambda t} \xi = \mathbf{u} + \mathbf{v}i$. $\Rightarrow$ Independent sol: $\mathbf{u}, \mathbf{v} \Rightarrow$ General solution: $\mathbf{y} = c_1 \mathbf{u} + c_2 \mathbf{v} + \dots$

AM>GM: (ctd.) (For each ξ_λ , under λ) $\mathbf{y}^{(1)} = e^{\lambda t} \xi_\lambda$ $\mathbf{y}^{(2)} = e^{\lambda t} (t \xi_\lambda + \eta_1)$ $\mathbf{y}^{(3)} = e^{\lambda t} (\frac{t^2}{2!} \xi_\lambda + t \eta_2 + \eta_1)$ $\dots$ $\mathbf{y}^{(n)} = e^{\lambda t} [\frac{t^{n-1}}{(n-1)!} \xi_\lambda + \sum_{i=0}^{n-2} \frac{t^i}{(i)!} \eta_{i+1}]$

2.1.2 Matrix Method | Exponential

Matrix Method: $\mathbf{y}(t) = e^{At} \mathbf{y}(0)$ (分两种方法: 1. 直接展开; 2. 通过一些已有 property 计算. ps: $e^{At} = \Psi$)

· **Property|AM=GM:** $e^{At} = T e^{Dt} T^{-1}$, $T = [\xi_1, \dots, \xi_n]$, $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, $e^{Dt} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t})$

· **Property|AM>GM:** $e^{At} = T e^{Jt} T^{-1}$, $T = [\lambda_1, \dots, \lambda_i, \eta_1^{(i)}, \dots, \lambda_j, \eta_j^{(j)}, \dots]$, $J = T^{-1} A T$ (Jordan form), $e^{Jit} = e^{\lambda_i t}$

Diagonalisation Method: (Ctd.) Let $\mathbf{y} = T\mathbf{x} \Rightarrow \mathbf{x}' = J\mathbf{x}$ [or $\mathbf{x}' = D\mathbf{x}$] 按前面方法解

Exponential of a matrix: $e^{At} := \sum_{n=0}^{\infty} \frac{(At)^n}{n!} = \mathbb{I} + At + \frac{1}{2!} A^2 t^2 + \dots = \lim_{n \rightarrow \infty} (\mathbb{I} + \frac{At}{n})^n$

· **Property:** $e^{A+B} = e^A e^B$ if $AB = BA$ || $e^{A \cdot 0} = \mathbb{I}$ || $[e^{At}]' = A e^{At}$

2.1.3 Matrix Method | Fundamental Matrix

Fundamental matrix Ψ : If $\{\mathbf{y}^{(j)}(t)\}$ is FSoS. $\Rightarrow \Psi(t) = [\mathbf{y}^{(1)}(t), \dots, \mathbf{y}^{(n)}(t)]$.

· **Property:** $\Psi'(t) = P(t)\Psi(t)$ (It's a sol). || $\det[\Psi(t)] = W(t) \neq 0$ || general solution: $\mathbf{y} = \Psi \mathbf{c}$, $\mathbf{c} = \Psi^{-1}(t_0) \mathbf{y}_0$

· **Relation with e^{At} :** $e^{At} = \Psi(t)$ with $\Psi(0) = \mathbb{I}$ || $e^{At} = \Psi(t) \Psi^{-1}(0)$

2.2 non-Homogeneous

Theorem: Solution of $(\star)$: $\mathbf{y} = \sum_i c_i \mathbf{y}^{(i)} + \mathbf{y}_p$, where $\mathbf{y}^{(i)}$ is solution of it's Homogeneous. 通常会让特解中积分后的常数 c=0

Variation of Parameters: $\mathbf{y} = \mathbf{y}_g + \mathbf{y}_p$; $\mathbf{y}_g = \Psi(t)\Psi^{-1}(t_0)\mathbf{y}(t_0)$; $\mathbf{y}_p = \Psi(t)\mathbf{u}(t)$, $\mathbf{u}(t) = \int_{t_0}^t \Psi^{-1}(s)\mathbf{g}(s)ds$. 经常 $\Psi(t)\mathbf{u}' = \mathbf{g}$ 来找 $\mathbf{u}$

Matrix methods|Diagonalisation: ¹Let $\mathbf{y} = T\mathbf{x}$ ² $(\star) \Rightarrow \mathbf{x}' = D\mathbf{x} + T^{-1}\mathbf{g} = D\mathbf{x} + \mathbf{h}$ ³直接计算: $\mathbf{z}_i = c_i e^{r_i t} + e^{r_i t} \int_{t_0}^t e^{-r_i s} h_i(s) ds$ $\mathbf{y}_p$

Undetermined Coefficients: when $\mathbf{g}(t)$ is polynomials, or real/complex exponential. Let $P_n(t) = \sum_{i=0}^n \mathbf{p}_i t^i$, $A_n(t) = \sum_{i=0}^n \mathbf{a}_i t^i$

¹构造 $\mathbf{y}_p$ 与 $\mathbf{g}$ 结构相匹配.(e.g. If $\mathbf{g} = P_n(t) \Rightarrow \mathbf{y}_p = A_n(t)$; If $\mathbf{g} = \mathbf{c}e^{kt} \Rightarrow \mathbf{y}_p = \mathbf{a}e^{kt}$) ²特殊: If $\mathbf{g} = \mathbf{c}e^{\lambda t}$, λ is eigenvalue with $AM = n \Rightarrow \mathbf{y}_p = e^{\lambda t} \sum_{i=0}^n \mathbf{a}_i t^i$ ³带入原方程得未知系数

2.3 Geometrical Interpretation & Linear Approximation

* 此节中, Consider nonlinear autonomous system: $\mathbf{x}' = F(x, y), \mathbf{y}' = G(x, y) (**)$

Phase Plane: O_{xy} referred as phase plane. **Trajectories:** Sol of ODE. $\mathbf{x}(t) = (x(t), y(t))^T$ direction $(dx/dt, dy/dt)$ **Phases Portrait:** 一组有代表性的轨迹

Critical Point (Equilibrium Point): $\mathbf{x}_0 = (x_0, y_0)$ is critical point iff $[x' = F(\mathbf{x}_0) = 0, y' = G(\mathbf{x}_0) = 0]$ or $[\mathbf{x}' = \mathbf{0}]$ (critical point 外变化简单)

Stable: Critical Point $\mathbf{x}_0$ is stable iff $\forall \epsilon > 0, \exists \delta > 0$ s.t. $\forall$ sols $\mathbf{x}(t)$ with $\|\mathbf{x}(0) - \mathbf{x}_0\| < \delta$, then $\|\mathbf{x}(t) - \mathbf{x}_0\| < \epsilon, \forall t \geq 0$

· 直观理解: 一开始离 critical point 很近, 那么随着时间推移也会保持在 critical point 附近. ps:attrcting 是会趋近于 critical point.

Attracting: Critical point $\mathbf{x}_0$ is attracting iff $\exists \delta > 0$ s.t. $\forall$ sols $\mathbf{x}(t)$ with $\|\mathbf{x}(0) - \mathbf{x}_0\| < \delta$ satisfies: $\lim_{t \rightarrow \infty} \mathbf{x}(t) = \mathbf{x}_0$

Asymptotically Stable: Critical point $\mathbf{x}_0$ is asy.stab iff ¹ Stable, ² Attracting.

Theorem: If $\mathbf{x}_*(t) = (x_*, y_*)$ s.t. $F(\mathbf{x}_*), G(\mathbf{x}_*) \neq 0. \Rightarrow \exists H$ s.t. $H(x, y) = (\tilde{x}, \tilde{y}), \tilde{x}' = 1, \tilde{y}' = 0$ H is a smooth change of variable function in neighbourhood of $\mathbf{x}_*$

Non-Linear $\rightarrow$ Linear: For system $(**)$, assume $\mathbf{x}_0 = (x_0, y_0)$ is critical point. $\Rightarrow \mathbf{u} = \mathbf{x} - \mathbf{x}_0$, then $\mathbf{u}' \approx A\mathbf{u}$ Linear; $A = \begin{pmatrix} \partial_x F(\mathbf{x}_0) & \partial_y F(\mathbf{x}_0) \\ \partial_x G(\mathbf{x}_0) & \partial_y G(\mathbf{x}_0) \end{pmatrix}$ Jacobian

2.4 Classification of Critical Points

* 此节中, Consider linear system: $\mathbf{x}' = A\mathbf{x}$ Critical Point(s): when $\mathbf{x}' = A\mathbf{x} = \mathbf{0}$

Classification of Critical Points at (x_0, y_0) (Linear): Let r_1, r_2 be sol of $\det(A - \lambda I) = 0$. · $\mathbb{C} : r = \lambda \pm i\mu (\mu > 0)$

· If A constant, write sol: $\mathbf{x} = c_1 e^{r_1 t} \xi_1 + c_2 e^{r_2 t} \xi_2$ · $GM = 1: \mathbf{x} = c_1 e^{rt} \xi + c_2 e^{rt} (t\xi + \eta)$ · AL: Almost Linear Systems.

$\mathbb{R}/\mathbb{C}$	Condition Stability	Type Name	Phase Plane Description	Other
$\mathbb{R}$	$r_1 < r_2 < 0$ asy.stab	N NSk	向原点, ξ_2 直线, ξ_1 曲线, and ξ_1 周围 $y = \pm x^3$	$c_2 \neq 0, t \rightarrow \infty: \xi_2$ 主导方向; $c_2 = 0, t \rightarrow \infty: \xi_1$ 主导方向
	$r_1 > r_2 > 0$ unstable	N NSo	原点向外, ξ_2 直线, ξ_1 曲线, and ξ_1 周围 $y = \pm x^3$	$c_1 \neq 0, t \rightarrow \infty: \xi_1$ 主导方向; $c_1 = 0, t \rightarrow \infty: \xi_2$ 主导方向
	$r_1 > 0 > r_2$ unstable	SP SP	$t \rightarrow \infty, \xi_1$ 从原点向外, ξ_2 从外向原点 and: 像 $y = \pm \frac{1}{x}$, 同进同出	$t \rightarrow \pm \infty : \mathbf{x} \rightarrow \infty$; $t \rightarrow \infty : c_1, c_2 \neq 0, \mathbf{x} \rightarrow \infty, \xi_1$ 主导; $t \rightarrow \infty : c_2 = 0, \mathbf{x} \rightarrow \infty, \xi_1$ 主导; $t \rightarrow \infty : c_1 = 0, \mathbf{x} \rightarrow 0, \xi_2$ 主导
	$r_1 = r_2 < 0$, GM=2 asy.stab	PN PN or Stable Star	直线 向原点	直线, u_1/u_2 is t independent
	$r_1 = r_2 > 0$, GM=2 unstable	PN PN or Unstable Star	直线 从原点向外	直线, u_1/u_2 is t independent
	$r_1 = r_2 < 0$, GM=1 asy.stab	IN (AL:Type: SpP) IN (Stable)	S 曲线, 向原点	$t \rightarrow \infty, \mathbf{x} \rightarrow 0, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2 \xi$ 方向相同
	$r_1 = r_2 > 0$, GM=1 unstable	IN (AL:Type: SpP) IN (Unstable)	S 曲线, 从原点向外	$t \rightarrow \infty, \mathbf{x} \rightarrow \infty, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2 \xi$ 方向相同
$\mathbb{C}$	$\lambda \neq 0, \lambda > 0$ unstable	SpP Unstable Focus	向外椭圆 (elliptical) 螺旋	$t \rightarrow \infty, \mathbf{x} \rightarrow \infty$ ps: 考虑 $A = (a, b; c, d)$, 如果 bc>0, 顺时针, 如果 bc<0, 逆时针
	$\lambda \neq 0, \lambda < 0$ asy.stab	SpP Stable Focus	向内椭圆 (elliptical) 螺旋	$t \rightarrow \infty, \mathbf{x} \rightarrow 0$ ps: 考虑 $A = (a, b; c, d)$, 如果 bc>0, 顺时针, 如果 bc<0, 逆时针
	$\lambda = 0$ stable (AL:Indeterminate)	C (AL:C or SpP) C	椭圆 (elliptical) and 半长轴 ξ 实部方向	Bounded trajectory or $\exists$ Periodic Trajectories
ps: N = Node PN = Proper Node IN = Improper / Degenerate Node SP = Saddle Point SpP = spiral (focus) point C = Center NSk = Nodal Sink NSo = Nodal Source				
For Almost Linear systems (见 2.5): $\mathbf{u}' = A\mathbf{u} + \underline{\eta}$, 如果 linear Approximation (去掉 $\underline{\eta}$), 按照 Linear 考虑. 如果不去 $\underline{\eta}$, 按照去掉的算 r_1, r_2 , Type 按照正常 / (AL) 的来.(图像会到 (0, 0))				

Some Defs: *Separatrix*: 将不同区域分开的轨迹; *Basin of attraction*: 所有趋近 critical point 的轨迹. ps: In any nonlinear ODEs, it's important to determine the basin of attraction.

3 Non-Linear ODE Methods

* 此节中, Consider two-dimensional autonomous system: $\mathbf{x}' = F(x, y), \mathbf{y}' = G(x, y)$ and $\mathbf{x} = (x, y)^T$ (#) Lyapunov's: 不需要知道微分方程的确切解 critical value. Poincaré-Bendixson: 寻找 limit cycle

Positive (Negative) [semi-]definite: Function $E : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is positive (negative) [semi-]definite iff: ps: positive [semi-]definite 正 (半) 定

$^1 E(\mathbf{x}_0) = 0, ^2 \forall \mathbf{x} \in D, \mathbf{x} \neq \mathbf{x}_0, E(x, y) > 0$ or $(E(x, y) < 0)$ or $[E(x, y) \geq 0]$ ps: Neighbourhood 领域

Isolated Critical Point: Critical Point $\mathbf{x}_0$ isolated iff $\exists$ Neighbourhood containing no other critical points. $\Downarrow E$: Lyapunov function

Lyapunov Stability Theorem: ¹(#); ²ICP $\mathbf{x}_0 \in D$; ³ $\exists$ 正定且连续 $E : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}, E(\mathbf{x}_0) = 0$; $\parallel ^4 \frac{dE}{dt} = \nabla E \cdot \mathbf{x}' = \partial_x E \cdot F + \partial_y E \cdot G$ 负 (半) 定 on $D \setminus \mathbf{x}_0 (D) \Rightarrow \mathbf{x}_0$ asp.(stab)

· **Look for function E :** Way 1: May be energy, but not always. Way 2: e.g. $E = ax^{2n} + by^{2m}, E = a(x + y)^{2n} + by^{2m}$ ps: $\frac{dE}{dt} = \nabla E \cdot \mathbf{x}'$

Lyapunov Instability Theorem: ¹(#); ²ICP $\mathbf{x}_0 \in D$; ³ $\exists E : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}, \partial_t E$ 连续, $E(\mathbf{x}_0) = 0$; ⁴ $\forall$ 领域 $\mathbf{x}_0, \exists \mathbf{x}_1 E(\mathbf{x}_1) > 0$; $\parallel ^5 \frac{dE}{dt}$ 正定 on $D \Rightarrow \mathbf{x}_0$ unstable

Periodic Solution: Soln $\mathbf{x}$ has: $\mathbf{x}(t + T) = \mathbf{x}(t), \forall x$. $\parallel$ 闭合曲线 $\Leftrightarrow$ Periodic sol $\parallel$ Periodic Solution + 附近有吸引或排斥 $\Rightarrow$ Limit Cycle $\parallel$ when $\dot{r} = 0, \dot{\theta} = const$. Periodic sol

Limit cycle: Limit cycle $\Rightarrow$ Periodic Solution $\parallel$ Def $t \rightarrow \pm \infty, \exists$ non-closed trajectory asymptotes to it. (inside or outside) ps: It's Orbital Stability

1. Limit cycle is: ¹asy.stab: $\forall$ trajectories asy. it; ²semistable: $\exists$ trajectories 单方向进/出; ³unstable: $\forall$ trajectories away from it.
2. **Theorem (If $\exists$ limit cycle / closed):** ¹ F, G 一阶偏导连续 in simply connected D ; ² $\exists$ 极限环/封闭环 $(\star)$ in $D. \Rightarrow ^1 \exists$ CP in $(\star)$; ²If $\exists!$ CP in $(\star)$, it's not SP.
3. **Theorem (No limit cycle / closed):** ¹ F, G 一阶偏导连续 in simply connected D ; ² $\partial_x F + \partial_y G$ 符号不变 on $D \Rightarrow$ No closed trajectory in D .
2 的推论: ¹If simple connected D no CP $\Rightarrow$ No closed trajectory in D . ²If simple connected D has unique CP $\Rightarrow$ No closed trajectory in D
- Poincaré-Bendixson Theorem:** ¹ F, G 一阶偏导连续 in D ; ² $\exists D_1 \subset D$ bounded; ³ $R = D_1 \cup \partial D_1$ has NO CP; $\parallel ^4 \exists$ soln $\mathbf{x} \in R, \forall t \geq t_0 \Rightarrow$
 $\Rightarrow ^1 \exists$ closed trajectory. ² Either $\mathbf{x}$ spirial towards a Periodic trajectory. or $\mathbf{x}$ periodic ³ R is not simply connected
- **Find Trapping Region:** trapping region: 符合上面定理 2-4 除 CP 的 region. Method: Consider polar coordinate, 计算 $r^2 = x^2 + y^2$ 对 t 的求导, 寻找 $rr' \geq 0$ or $rr' \leq 0$

4 Fourier Series

* BVPs 可能没有解/一个解/无数解; IVP 只会会有一个解

4.1 Function Space, Eigenfunctions, Periodic Function and Orthogonal Function

Periodic Functions: Function $f(x)$ is Periodic iff $f(x+T) = f(x), \forall x$. Period: T. Fundamental Period: The smallest possible T.
· Property: If f, g has same period T . $\Rightarrow af(x) + bg(x)$ also has period T . || For $\sin(\omega x), \cos(\omega x): T = 2\pi/\omega$
Differential Operators: Any linear combination of n -th derivatives defines a linear map or linear operator. e.g. $p(x) \frac{d}{dx} + \frac{d^2}{dx^2} : y \mapsto p(x) \frac{dy}{dx} + \frac{d^2y}{dx^2}$
Eigenvalues & Eigenfunctions: If D is a differential operator $\Rightarrow$ If $Df = \lambda f$, then λ is Eigenvalue and f is Eigenfunctions
· Property: 对于微分方程: $Df = \lambda f \Rightarrow$ solution: $f = \sum c_n f_n$ f_n are Eigenfunctions
Inner Product: $(f, g) = \int_a^b f(x) \overline{g(x)} dx$ For funcs f, g in (a, b) || Orthogonal: $(f, g) = 0$ || Square Norm: (f, f) .
Kronecker delta: $\delta_{mn} = \{0, m \neq n; 1, m = n$.

4.2 Fourier Series

Fourier Series: Periodic function with $T = 2L$, $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L}))$ (*) where a_0, a_n, b_n are Fourier Coefficients.
Euler-Fourier Formulas: Suppose (*) converges in the interval $[-L, L]$. Then:
 $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos(\frac{n\pi x}{L}) dx$ $b_n = \frac{1}{L} \int_{-L}^L f(x) \sin(\frac{n\pi x}{L}) dx$ $a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$ (Average Value of f over $2L$)
Piecewise Continuous: $f(x)$ piecewise continuous on $[a, b]$ iff $\exists a = x_0 < \dots < x_n = b$ s.t. $^1 f(x)$ continuous $\forall x \in (x_{i-1}, x_i)$ $^2 f(x_i^{\pm}) = \lim_{x \rightarrow x_i^{\pm}} f(x)$ finite.
Fourier Convergence Theorem: If $^1 f, f'$ piecewise continuous on $[-L, L]$, $^2 f(x+2L) = f(x)$. $\Rightarrow$ Fourier Series of f converges.
· Furthermore, Let Fourier Series of f is F . $F(x) \rightarrow f(x), f(x)$ continuous; $F(x) \rightarrow \frac{f(x^+) + f(x^-)}{2}, f(x)$ discontinuous.
Theorem: Fourier Series for piecewise continuous functions can be Integrated term-by-term to Convergent Fourier Series.
Finite Truncation (Partial Sum): Finite Truncation of Fourier Series is: $s_k = \frac{a_0}{2} + \sum_{n=1}^k (a_n \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L}))$ || Error: $e_N = s_N - f(x)$
· $n \rightarrow \infty$: 1 Continuous points, Fourier approximation of $f(x)$ is better ($|e_n|$ decrease). 2 Discontinuous points, oscillation of Fourier series get smaller.
· Gibbs Phenomenon: The Amplitude of the largest oscillation (near discontinuous points) does not decrease.

4.3 Other Results on Fourier Series

Extension of a function: $f(x)$ know for $x \in [0, L] \Rightarrow$ Even Ext.: $h(x) = \{f(x), [0, L]; f(-x), (-L, 0)\}$ || Odd Ext.: $h(x) = \{f(x), (0, L); 0, x = \pm L, 0; -f(-x), (-L, 0)\}$
Even Extension: $a_n = \frac{2}{L} \int_0^L f(x) \cos(\frac{n\pi x}{L}) dx, b_n = 0$ || Odd Extension: $b_n = \frac{2}{L} \int_0^L f(x) \sin(\frac{n\pi x}{L}) dx, a_n = 0$
Complex Form of Fourier Series: $f(x) = \sum_{-\infty}^{\infty} c_n e^{\frac{in\pi x}{L}}$ || $c_n = \frac{a_n - ib_n}{2}; c_{-n} = \frac{a_n + ib_n}{2}; c_0 = \frac{a_0}{2}, n > 0$ || $a_0 = 2c_0; a_n = c_n + c_{-n}; b_n = i(c_n - c_{-n})$
Euler-Fourier Formula (Complex): If $(\uparrow)$ of Periodic $f, (T = 2L)$ converges in $[-L, L] \Rightarrow c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-\frac{in\pi x}{L}} dx$ || $c_{-n} = \overline{c_n}$ if f real.
Parseval's Theorem: If 1 Periodic function $f; ^2 f$ has convergent Fourier Series.
 $\Rightarrow$ Square Norm $(f, f) = \int_{-L}^L |f(x)|^2 dx = 2L \sum_{-\infty}^{\infty} |c_n|^2 = L [\frac{|a_0|^2}{2} + \sum_{n=1}^{\infty} (|a_n|^2 + |b_n|^2)]$ ps: 无限维的勾股定理

5 Linear PDEs by Separation of Variables

Requirement: Assume $u(x, t) = X(x)T(t)!$ $\Rightarrow$ If PDE 能写成类似 $\frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} \Rightarrow \frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} = \lambda$ (Seperation Constant) $\Rightarrow$ 可以使用此方法!
1.Homo, 假设: $u(0, t) = u(L, t) = 0; u(x, 0) = f(x)$ 2.Inhomo, 假设: $u(0, t) = a, u(L, t) = b; u(x, 0) = f(x)$
Basic Idea|Homo: 1. 假设解 u 能写成 $u(x, t) = X(x)T(t) \Rightarrow$ 2. 代入 PDE, 得两个 ODE $\Rightarrow$ 3. 解其中一个得到特征函数 $X_n \Rightarrow$ 4. 代 initial 条件得限制条件 $\Rightarrow$
5. 解另一个得到特征函数 $T_n \Rightarrow$ 6. 特征函数 $X_n T_n$ linear combination 得到 u . 7. 代 boundary 条件用 Euler-Fourier 得到的系数.
· Remark: 1 化简中可以使用正比简化计算 2 对 n 的取值为 $1, 2, 3, \dots$ 3 对 λ 的取值让出现 $\sin, \cos$ 时为正不容易出错, 或注意正负号
Basic Idea|Inhomo: 1. Let $u(x, t) = v(x) + \omega(x, t), v(x) = \frac{b-a}{L}x + a \Rightarrow \omega(x, t)$ satisfy: $\omega(x, 0) = f(x) - v(x); \omega(0, t) = \omega(L, t) = 0$ 2. 按 Homo 解 $\omega(x, t)$
Laplace Equation: 2D: $\nabla^2 u = \partial_x^2 u + \partial_y^2 u = 0$ 3D: $\nabla^2 u = \partial_x^2 u + \partial_y^2 u + \partial_z^2 u = 0$ Following are the method to solve:
1. Rectangle: $(0 \leq x \leq a, 0 \leq y \leq b)$ with $u(x, 0) = u(x, b) = u(0, y) = 0; u(a, y) = f(y)$ 直接按照 Homo 正常计算 $\Downarrow$ 可能用到 Euler Equation
2. Circle: $(u(r = a, \theta) = f(\theta); u(r, \theta) \text{ bounded})$ $^1 \partial_x^2 u + \partial_y^2 u = 0; x = r \cos(\theta), y = r \sin(\theta)$ 变极坐标 $R(r), \Theta(\theta)$ 2 一样的计算 + 条件 $\Theta(\theta + 2\pi) = \Theta(\theta)$

6 Sturm-Liouville Theory

7 Appendix

Trigonometric func: $s+s=cs, s-s=cs, c+c=cc, c-c=-ss$ || e.g. $\sin A + \sin B = 2\sin(\frac{A+B}{2})\cos(\frac{A-B}{2})$ $\sin A \sin B = -\frac{1}{2}[\cos(A+B) - \cos(A-B)]$
· $\sin(2x) = 2\sin(x)\cos(x)$ $\cos(2x) = 2\cos^2(x) - 1 = 1 - \sin^2(x) = \cos^2(x) - \sin^2(x)$ $\cos(x) = \sqrt{\frac{1+\cos(2x)}{2}}$ $\sin(x) = \sqrt{\frac{1-\cos(2x)}{2}}$
hyperbolic func: $\sinh(x) = \frac{e^x - e^{-x}}{2}$ $\cosh(x) = \frac{e^x + e^{-x}}{2}$ $\cosh^2(x) - \sinh^2(x) = 1$ $[\sinh(x)]' = \cosh(x)$ $[\cosh(x)]' = \sinh(x)$
· $\sinh(x \pm y) = \sinh(x) \cosh(y) \pm \cosh(x) \sinh(y)$ $\cosh(x \pm y) = \cosh(x) \cosh(y) \pm \sinh(x) \sinh(y)$
Common Int: $\int \tan(x) dx = -\ln|\cos(x)|; \int \cot(x) dx = \ln|\sin(x)|; \int \sec(x) dx = \ln|\sec(x) + \tan(x)|; \int \csc(x) dx = \ln|\csc(x) - \cot(x)| = \ln|\tan(\frac{x}{2})|$
Common Der: $[\tan(x)]' = \sec^2(x); [\cot(x)]' = -\csc^2(x); [\sec(x)]' = \sec(x)\tan(x); [\csc(x)]' = -\csc(x)\cot(x); [\arctan(x)]' = \frac{1}{1+x^2}$
· $1 + \tan^2(x) = \sec^2(x); 1 - \csc^2(x) = \cot^2(x); \csc^2(x) - \cot^2(x) = 1$. $[\arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}; [\arccos(x)]' = -\frac{1}{\sqrt{1-x^2}}$
Polar Coordinate: $x = r \cos(\theta), y = r \sin(\theta)$ we have: $r\dot{r} = x\dot{x} + y\dot{y}, y\dot{x} - x\dot{y} = -r^2\dot{\theta}$
Explain of Fourier Series: In application, it's decomposition of signal into infinite (countable) waves; each wave $\cos(\frac{n\pi x}{L}), \sin(\frac{n\pi x}{L})$ has discrete period; a_n, b_m correspond to amplitudes of waves. || In vector algebra, it's a space of periodic functions; f corresponds to an arbitrary vector v ; orthogonal basis $\{1, \cos(\frac{n\pi x}{L}), \sin(\frac{n\pi x}{L})\}; a_n, b_m$ correspond to coefficient of each components of basis.
Even & Odd Function: Even: $f(-x) = f(x)$; Odd: $f(-x) = -f(x)$. || Property: $^1 f(x)$ odd: $\int_{-L}^L f(x) dx = 0$ $^2 f(x)$ even: $\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx$
· Other Property: 1 even + even = even; 2 odd + odd = odd; 3 even $\times$ even = even; 4 odd $\times$ odd = even; 5 even $\times$ odd = odd.

Euler Formula: $e^{ix} = \cos(x) + i\sin(x)$ || $e^{inx} = \cos(nx) + i\sin(nx)$ || $\cos(nx) = \frac{e^{inx} + e^{-inx}}{2}$ || $\sin(nx) = \frac{e^{inx} - e^{-inx}}{2i}$
Slove $y'' + \lambda y = 0$: ¹ $\lambda < 0 : y = c_1 \sinh(\sqrt{-\lambda}x) + c_2 \cosh(\sqrt{-\lambda}x)$ ² $\lambda = 0 : y = c_1x + c_2$ ³ $\lambda > 0 : y = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$
Approximation $\sin(x) \approx x$ when $x \rightarrow 0$